PROJECT REPORT

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Project Information	Details
Project Title	Global Olympic Sports Performance Analysis
Project Domain	Data Analytics and Visualization
Primary Tool	Tableau Public / Tableau Desktop (confirm version used)
Web Integration	Python Flask with embedded Tableau dashboards (planned / verify implementation)
Dataset	athlete_events.csv
Dataset Coverage	Olympic records from 1896 to 2016; Summer and Winter Games
Team ID	_____
Team Members	1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____
Institution / Department	_____
Submission Date	_____

Prepared as a project report using the supplied project description and report-format template.

Note: Replace blank team and institutional details. Insert actual Tableau screenshots and working project links before final submission.

1. INTRODUCTION

1.1 Project Overview

The Global Olympic Sports Performance Analysis project uses historical Olympic athlete-event data to explore medal performance across countries, sports, seasons and years. The project is designed around three stakeholder scenarios: a national performance strategist, a sports investment analyst and an Olympic data storyteller. Tableau is used to create interactive visualizations, dashboards and a story that make the historical data easier to explore.

The supplied athlete_events.csv dataset is described as containing 271,116 rows and 15 columns, covering Olympic records from 1896 to 2016 and including both Summer and Winter Games. The available fields include athlete identity and attributes, team and National Olympic Committee (NOC), Games, Year, Season, City, Sport, Event and Medal.

1.2 Purpose

The purpose is to convert a large historical dataset into understandable visual evidence that can help users:

- Compare medal-winning performance across countries and sports.
- Explore medal records relative to athlete participation.
- Identify long-term medal trends across Olympic editions and seasons.
- Examine host-nation performance using a verified host-country lookup.
- Explore investment and regional efficiency when reliable supplementary data is available.

The project is an exploratory historical analysis. It does not prove that investment causes medal success, and its current source file cannot provide results after 2016.

2. IDEATION PHASE

2.1 Problem Statement

Olympic results are spread across many years, sports, events and participating countries. A user who wants to compare national performance or identify sport-level patterns may find it difficult to interpret the raw athlete-event records directly. The project addresses this problem by organizing the data into calculated measures and interactive Tableau views.

A second problem is data completeness: the supplied CSV does not contain sports investment, region or a dedicated host-country field, and it ends in 2016. Therefore, some questions in the project scenarios require verified external lookup tables or additional datasets. These gaps must be stated rather than filled with invented values.

2.2 Empathy Map Canvas

The following empathy map is based on the three stakeholder scenarios provided in the project brief. It represents assumed information needs for design purposes; it is not the result of recorded interviews.

Stakeholder	Thinks / Needs	Sees / Uses	Pain Points	Desired Outco
Priya – Olympic Performance Strategist	Which countries and sports perform strongly? Where should attention be focused?	Country medal comparisons, sport medal distribution and medals per athlete.	Raw tables are difficult to compare; medal counts can be misleading if team members are counted as separate medals.	Clear country a support evidenc
Carlos – Sports Investment Analyst	Do higher investments correspond with better medal outcomes? Which regions appear efficient?	Investment amounts, country medal outcomes, region and athlete participation.	The supplied CSV has no investment or region fields.	A defensible inv view using verif
William – Olympic Data Storyteller	How have medals changed over time? Do host nations show different performance?	Historical trends, host-country medal comparisons and regional medal distribution.	Long time ranges and host information need careful data preparation.	A clear historica interactive char

2.3 Brainstorming

The project ideas were grouped into three dashboard themes:

Theme	Ideas Selected	Planned Output
National Performance	Medals per athlete by sport; country medal ranking; medal distribution by sport.	Dashboard 1 – Global Olympic Performance Analys

Theme	Ideas Selected	Planned Output
Investment & Regional Efficiency	Investment versus medal outcomes; regional efficiency; country-wise efficiency.	Dashboard 2 – Olympic Investment and Regional E external investment and region data.
Trends & Storytelling	Medal trends by year and season; host-nation comparison; regional medal distribution by sport.	Dashboard 3 – Historical Olympic Trends and Host four-point Tableau Story.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Journey Stage	User Activity	User Question	System Response / Output
Discover	Open the project overview and dataset summary.	What does this project analyze?	Show project purpose, dataset range, fields
Explore	Select year, season, country or sport filters.	Which countries and sports have the most medals?	Update country, sport and athlete-level chart
Compare	Move between national and regional dashboards.	How do outcomes differ by country or region?	Show medal totals and efficiency measures; region mapping.
Investigate	Inspect trends and host-country views.	How have results changed over time? What happened when a nation hosted?	Show medal trend and host comparison usin
Present	Navigate the Tableau Story and Flask page.	What are the main findings and limitations?	Present the dashboards in sequence and sta dependencies.

3.2 Solution Requirement

Requirement Type	Requirement
Data	Load athlete_events.csv and verify fields, data types, row count and year coverage.
Data preparation	Handle missing Medal values; create medal count and a unique medal key for distinct medal-event counting.
Calculated measures	Create medals per athlete and project-defined medal efficiency.
Visualizations	Create nine visualizations covering national performance, investment / regional efficiency and historical trends.
Dashboards	Combine the nine visualizations into three interactive dashboards.
Story	Create four Tableau Story points: overview and one point for each scenario dashboard.
Interactivity	Provide Year, Season, NOC, Sport and Medal filters; Region filter after region mapping is added.
Web integration	Publish Tableau content and embed it in a Flask website using a supported Tableau embedding method.
Testing	Validate calculations, filters, data scope, loading behaviour and embedded dashboard access.

3.3 Data Flow Diagram

Logical data flow for the proposed solution:

Stage	Input / Process	Output
1. Source	athlete_events.csv	Raw Olympic athlete-event records.
2. Preparation	Check types and missing values; verify Year, Season, NOC, Sport, Event and Medal fields.	Cleaned / validated dataset.
3. Calculations	Medal Count; Unique Medal Key; Medal Efficiency; Medals per Athlete.	Measures for Tableau analysis.
4. Supplementary data	Verified investment table, region-to-NOC mapping and host-NOC lookup (where available).	Additional dimensions for scenarios 2 and 3.
5. Tableau	Build nine worksheets, three dashboards, filters and four story points.	Interactive analytical views.
6. Flask	Embed published Tableau dashboards / story in a Flask web page.	Browser-based project demonstration.

Data flow: CSV + verified lookup tables → data validation and calculated fields → Tableau worksheets → dashboards and story → published Tableau content → Flask web interface.

3.4 Technology Stack

Technology	Purpose in Project	Status / Note
CSV dataset	Stores historical athlete-event records.	Source dataset identified as athlete_ever
Tableau Public / Desktop	Data connection, calculated fields, charts, dashboards, filters and story.	Use the edition actually installed.
Python + Flask	Provides a lightweight web application and navigation around embedded Tableau views.	Implementation must be verified before c
HTML / CSS	Builds the Flask page layout and navigation.	Used for the web presentation layer.
GitHub	Stores project source code and documentation.	Insert the actual repository URL in Apper
Browser	Runs the Flask site and embedded Tableau content.	Requires access to the published Tablea

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the problem because Tableau can transform the raw athlete-event dataset into multiple linked views. Country bars answer ranking questions, sport charts reveal discipline-level patterns, line charts show changes over time, and dashboard filters let users narrow the analysis. The solution is appropriate for exploratory analysis and classroom demonstration.

The solution only partially fits the investment and regional scenario until additional verified data is connected. The dataset itself cannot support claims about funding, regions or post-2016 results.

4.2 Proposed Solution

No.	Visualization	Chart Type	Purpose / Question
1	Average Medals per Athlete by Sport	Horizontal bar	Which sports have the highest medal-winning athlete-record rate per distinct athlete
2	Global Olympic Performance by Country	Vertical bar	Which NOCs have the highest distinct medal-event counts?
3	Medal Distribution by Sport	Treemap	Which sports contribute the most distinct medal-event records?
4	Sports Investment vs Medal Performance	Scatter / bubble	How do verified investment values compare with medal outcomes?
5	Average Medal Efficiency by Region	Pie (bar recommended for comparison)	How do project-defined efficiency values compare across mapped regions?
6	Country-wise Medal Efficiency	Horizontal bar	Which countries have higher medal counts relative to participating athletes?
7	Trend of Total Medals Over the Years	Line	How do medal-event counts change across Olympic years and seasons?
8	Olympic Host Nations Medal Comparison	Vertical bar	How many medals did host NOCs win in their home Games?
9	Regional Medal Distribution by Sport	Donut	How are medal-event counts distributed across mapped regions for a selected spor

4.3 Solution Architecture

Layer	Components	Responsibility
Data layer	athlete_events.csv; optional investment, region and host-NOC lookup tables.	Provides raw records and supplementary dimensions.
Preparation layer	Field validation, missing-value handling and calculated fields.	Creates consistent measures and avoids counting team me medals where the key supports deduplication.
Analytics layer	Nine Tableau worksheets.	Implements the chart-level analysis.
Presentation layer	Three Tableau dashboards and four-point Tableau Story.	Groups charts by stakeholder scenario and provides guide
Web layer	Flask application, HTML/CSS and Tableau embed.	Provides navigation and access to published dashboards.

Architecture note: the exact Tableau publishing destination, Flask deployment environment and authentication method were not specified in the supplied project details. Confirm these from the actual implementation.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is organized into four proposed sprints. Each sprint is planned for 20 story points. The schedule below is a suggested sequence for this Olympic Tableau project; replace dates and completion values with the team's actual records.

Sprint	Main Work	Planned Story Points	Duration	Deliverable
Sprint 1	Dataset connection, field checks, data cleaning and calculated fields.	20	6 days	Validated data sourc
Sprint 2	Build visualizations 1–3 and Dashboard 1; validate country and sport metrics.	20	6 days	National Olympic Pe
Sprint 3	Build visualizations 4–6; prepare investment and region data dependencies.	20	6 days	Investment and Reg dashboard, subject
Sprint 4	Build visualizations 7–9, Dashboard 3, Tableau Story, Flask embedding and final testing.	20	6 days	Historical Trends da demo.

Suggested task breakdown:

- Backlog: dataset audit, cleaning, calculated fields, nine worksheets, three dashboards, story, Flask integration, testing and report.
- Dependencies: investment analysis needs investment data; regional views need NOC-to-region mapping; host analysis needs year / season / host-NOC mapping.
- Review each sprint using working Tableau sheets rather than relying only on a task-completion percentage.

Velocity reference: the supplied planning template illustrates 20 story points over a 10-day sprint as 2 story points per day. If the project uses six-day sprints, calculate velocity using the team's actual completed points and actual sprint duration; do not copy the example value as measured performance.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should verify both correctness and usability. The following test cases are proposed; record actual outcomes and timings after running them in the team's Tableau environment.

Test ID	Test Area	Test Procedure	Expected Result
T01	Data load	Connect athlete_events.csv and check the number of records and fields.	271,116 rows and 15 columns, as described fo
T02	Year range	Inspect minimum and maximum Year values.	Dataset coverage should end in 2016; do not di present.
T03	Medal Count	Check rows with Gold, Silver or Bronze and missing Medal values.	Only medal-winning athlete-event records recei
T04	Unique Medal Key	Compare distinct keys with sample events, especially team sports.	Team members for the same NOC / event / me as separate medals where the key is valid.
T05	Filters	Test Year, Season, NOC, Sport and Medal filters.	Charts update consistently and filter scope is in
T06	Supplementary data	Validate region, investment and host lookup relationships.	No duplicated rows or unmatched NOC / year /
T07	Dashboard performance	Open each dashboard and interact with filters.	Record load time and note slow worksheets or
T08	Flask integration	Open the Flask page and test each embedded Tableau view.	Published dashboards load and navigation wor

Performance measures to record: dashboard initial load time, filter response time, extract refresh time (if extracts are used), number of records loaded, and any calculation or relationship errors. Tableau Performance Recording may be used where available. No measured timings were provided, so this report does not invent performance results.

7. RESULTS

7.1 Output Screenshots

Insert actual screenshots from the completed Tableau workbook and Flask application in this section. The report template requests output screenshots, but no screenshots were supplied with the report-format file. Do not use illustrative images as evidence of a working implementation.

Figure	Screenshot to Insert	What It Should Show
Figure 1	Dashboard 1 – Global Olympic Performance Analysis	Sport-level medals per athlete, top 15 NOCs and sport medal distribution.
Figure 2	Dashboard 2 – Olympic Investment and Regional Efficiency	Investment scatter plot and regional / country efficiency views; include only if connected.
Figure 3	Dashboard 3 – Historical Olympic Trends and Host Performance	Medal trend, host-NOC comparison and regional medal distribution.
Figure 4	Tableau Story	Four story points: overview and three scenario dashboards.
Figure 5	Flask Web Application	Navigation and at least one successfully loaded embedded Tableau dashboa

Expected Analytical Outputs

The planned outputs are comparative rankings, sport contribution patterns, medal trends and host-country comparisons. Actual leading countries, leading sports and numerical findings must be taken from the workbook after filters and calculations are validated. The values mentioned in the project brief (for example, sample medal totals or sport averages) must not be treated as verified results unless reproduced from this exact dataset and calculation method.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
Interactive filters allow users to explore different years, seasons, NOCs and sports.	The source dataset ends in 2016 and cannot answer 2020–2026 quest
Three dashboards organize the analysis around distinct stakeholder needs.	Investment and region analysis cannot be completed from the CSV alo
Distinct medal keys can reduce over-counting of team-event medals.	The unique-key logic must be checked for event naming and relationsh
Tableau Story provides a guided presentation of the findings.	Pie and donut charts can make precise comparisons harder than sorte
Flask embedding can make dashboards accessible through a project website.	Published embeds depend on network access, Tableau permissions ar
The project can be extended with newer editions and trusted supplementary data.	Results depend on source quality, consistent NOC mappings and care

9. CONCLUSION

The Global Olympic Sports Performance Analysis project proposes a Tableau-based approach for exploring historical Olympic medal performance across countries, sports, seasons and years. Its planned deliverables are nine visualizations, three dashboards and a four-point Tableau Story, with Flask integration for a web-based demonstration.

The existing athlete_events.csv dataset supports historical medal and participation analysis through 2016. However, investment comparisons, regional comparisons and host-nation analysis require additional verified data. The project should clearly separate results produced directly from the CSV from results that depend on supplementary sources. The quality of the final outcome depends on validating medal-counting logic, avoiding duplicated records and testing all filters and embedded views.

10. FUTURE SCOPE

- Extend the dataset with verified Olympic editions after 2016, including the relevant Summer and Winter Games.
- Add a maintained NOC-to-region mapping to support regional comparisons.
- Add reliable country- and year-level sports investment data with documented currency, coverage and source definitions.
- Build a complete host-NOC lookup table for the Olympic editions included in the analysis.
- Improve performance using Tableau extracts, optimized data relationships and fewer unnecessary marks.
- Deploy the Flask application with suitable hosting, access control and secure configuration.
- Add more analysis such as medal trends by gender, age group, event category and athlete participation, where supported by the dataset.

11. APPENDIX

A. Source Code (if any)

Add the Flask application source code, HTML templates, CSS files and any data-preparation scripts here or provide a link to the repository. No source-code files were included with the report template, so this report does not claim to contain or reproduce the implementation.

B. Dataset Link

Dataset file: athlete_events.csv. Insert the actual dataset source / download URL used by the team:

Dataset URL: _____

C. GitHub & Project Demo Link

GitHub repository: _____

Tableau Public / Server dashboard link: _____

Flask project demo link: _____

D. Calculated Fields Reference

Calculated Field	Formula / Definition	Interpretation
Medal Count	IF NOT ISNULL([Medal]) AND [Medal] <> 'NA' THEN 1 ELSE 0 END	Counts medal-winning athlete-event records medals.
Unique Medal Key	IF [Medal Count] = 1 THEN STR([Year]) + '|' + [Season] + '|' + [Event] + '|' + [NOC] + '|' + [Medal] END	Creates a key intended to count a medal ev validate against source event structure.
Medal Efficiency	COUNTD([Unique Medal Key]) / COUNTD([ID]) * 100	Project-defined ratio of distinct medal keys t multiplied by 100; not an official Olympic me
Medals per Athlete	SUM([Medal Count]) / COUNTD([ID])	Medal-winning athlete records per distinct a

Important calculation note: the Medal Efficiency formula is a project-specific indicator. If the formula already multiplies by 100, format it as a number with a percent suffix rather than applying Tableau's percentage formatting again. Team-event medal counting and data relationships must be checked to prevent duplicate counts.